

Day 37 — Forecasting Recommendation

Recommended method

Based on the two holdout periods and the rolling-origin backtest, I recommend using **Prophet + Holidays** for the current forecasting system.

Its average rolling-origin MAE is approximately 87.19 sales, compared with approximately 132.09 sales for the seasonal naive baseline.

With typical daily sales around 3,300, an MAE of 200 sales would mean the forecast is typically within approximately 6% of actual sales.

When the model performs worst

The largest forecasting errors occur during unusual periods, especially the December/Christmas period. These periods contain sales behavior that cannot be fully inferred from the normal weekly and yearly patterns.

Is the simple baseline good enough?

The seasonal-naive baseline is a strong competitor because sales have a clear weekly cycle. If the selected model does not consistently beat the baseline in the rolling-origin backtest, I would prefer the seasonal naive forecast because it is simpler, easier to explain and less costly to maintain.

What information would improve the forecast?

The most useful additional information would be business events that cannot be inferred from historical sales alone, such as holidays, special promotions, pricing changes, stock availability and major store events.

Refitting strategy

I would refit the model at least weekly or whenever new daily data becomes available, depending on the operational forecasting schedule.

I would monitor the rolling MAE over time. If the model's MAE becomes consistently worse than the seasonal-naive baseline, this would be a sign that the model may have gone stale and should be investigated or retrained.

Final recommendation

The final deployment decision should be based on the rolling-origin backtest rather than a single holdout window. The model should only replace the seasonal-naive baseline if it provides a meaningful and consistent improvement.